IFEval Factor Analysis: Rank Selection, Sparsity Tuning, Predictive Comparison, and Selected-Model Interpretation

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1 Overview

This note summarizes the reproducible IFEval factor-analysis pipeline. The data are a binary model-by-item response matrix with 122 model rows and 500 IFEval item columns. The analysis has three goals:

1. select the working factor rank using held-out predictive performance;
2. tune the entrywise sparsity penalty on the loading matrix;
3. compare the selected independent-mixture probit factor model to an ordinary binary probit factor baseline;
4. interpret and visualize the selected factor space through loadings, cross-loadings, mixture profiles, and three-dimensional factor scores.

2 Models

The selected model is an independent marginal-mixture probit factor model,

$$X_{ij} \mid f_i, \alpha_j, \lambda_j \sim \text{Bernoulli}\{\Phi(\alpha_j + \lambda_j^\top f_i)\}, \tag{1}$$

$$f_{ih} \sim \sum_{g=1}^{G} \pi_{hg} N(\mu_{hg}, \sigma_{hg}^2), \qquad h = 1, \dots, H, \tag{2}$$

with independent mixture coordinates across factors. For the selected IFEval analysis, the working model uses $H = 3$ factors and $G = 3$ marginal mixture components.

The main comparison model is an ordinary binary probit factor model,

$$X_{ij} \mid f_i, \alpha_j, \lambda_j \sim \text{Bernoulli}\{\Phi(\alpha_j + \lambda_j^\top f_i)\}, \tag{3}$$

$$f_i \sim N(0, I_H), \tag{4}$$

also fit at $H = 3$ for the selected-model comparison. Because factor labels and signs are unidentified, the ordinary probit factors are aligned to the mixture factors by the signed permutation that maximizes factor-score correlations.

3 Rank Selection and Predictive Evaluation

Rank selection is based on held-out cell predictive log likelihood over $H = 1, \ldots, 10$. The comparison includes independent-mixture probit fits with $G = 2$ and $G = 3$, along with the ordinary Gaussian-factor probit baseline. All three methods select $H = 3$ by the best mean held-out log score, and the one-standard-error rule also selects $H = 3$.

Table 1: Selected rank and held-out log score by method.

Method	Best H	Mean log score	SE	Mean seconds at best H
Mixture probit, $G = 2$	3	-0.2762	0.0039	3.285
Mixture probit, $G = 3$	3	-0.2756	0.0033	3.434
Ordinary probit factor	3	-0.2792	0.0041	3.062

The $G = 3$ mixture has the strongest held-out log score, though the differences are modest relative to the fold-to-fold uncertainty. The main practical conclusion is therefore not only that $H = 3$ is selected, but also that the mixture model provides a slightly better predictive account without changing the selected rank.

Table 2: Held-out predictive metrics at $H = 3$.

Method	Accuracy	SE	Brier	SE	Log score	SE
Mixture $G = 2$	0.8861	0.0009	0.0824	0.0010	-0.2762	0.0039
Mixture $G = 3$	0.8866	0.0008	0.0822	0.0008	-0.2756	0.0033
Ordinary probit	0.8852	0.0017	0.0826	0.0011	-0.2792	0.0041

4 Lambda Sparsity Tuning

After selecting $H = 3$ and $G = 3$, the entrywise L1 penalty on the loading matrix is tuned over the grid $\{0, 0.5, 1, 2, 3, 4, 6, 8, 10, 12\}$. The tuning path uses a single common pretraining fit and then reruns the refinement stage for each candidate value. We summarize the tradeoff between unpenalized binary probit log likelihood and loading sparsity, and select the penalty by a BIC-style criterion in which the varying degrees of freedom are driven by the number of nonzero loading entries.

The raw likelihood favors less shrinkage, as expected, but the BIC-style criterion favors $\lambda_{\mathrm{L1}} = 10$. The criterion decreases through $\lambda_{\mathrm{L1}} = 8$, reaches its minimum at 10, and then increases at 12, so the expanded grid reaches the tuning shelf rather than selecting a boundary value. The selected penalty keeps about half of the loading entries nonzero in the tuning path and reduces the number of large entries with $|\lambda| > 0.5$ to 413.

5 Selected-Model Fit

The selected independent-mixture fit uses $H = 3$, $G = 3$, and the tuned loading penalty $\lambda_{\mathrm{L1}} = 10$. The fitted loading matrix has a clear asymmetry across axes: factor 1 is broad and dense, while factors 2 and 3 are more selective. Using a threshold $|\lambda_{jh}| \geq 0.5$, 349 of 500 items load on factor 1, compared with 30 items on factor 2 and 34 items on factor 3. This pattern supports reading

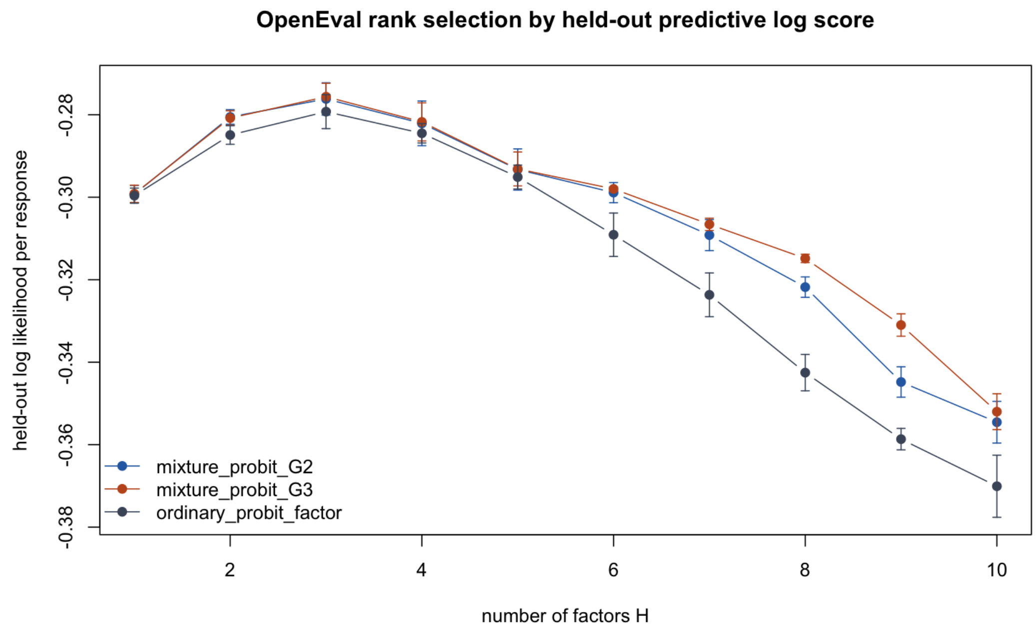

Figure 1: Held-out predictive log likelihood across candidate ranks.

factor 1 as a broad instruction-following or overall compliance axis, with factors 2 and 3 capturing more specialized modes of instruction control. The reported sparse loading matrix sets all entries below this threshold to zero and is saved as `results/loadings_crossloadings/ifeval_sparse_lambda_matrix_threshold_0p5.csv`.

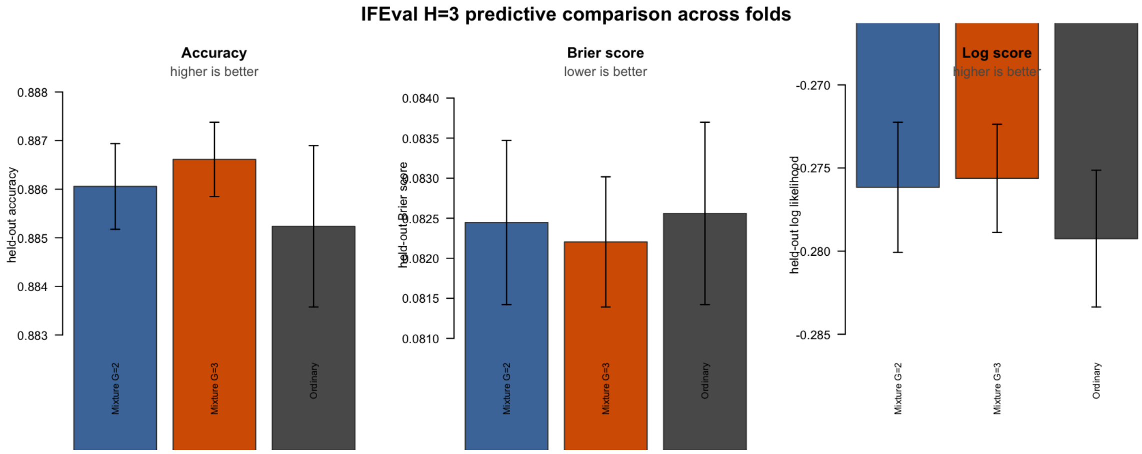

Figure 2: Held-out predictive metrics for the selected $H = 3$ fits.

Table 3: Entrywise loading-penalty tuning path for the selected $H = 3, G = 3$ model.

λ_{L1}	Log score	Nonzero loadings	Density	BIC-style score
0.0	-0.2166	1500	1.000	52762.9
0.5	-0.2185	1447	0.965	52402.0
1.0	-0.2202	1406	0.937	52158.0
2.0	-0.2237	1329	0.886	51740.0
3.0	-0.2274	1262	0.841	51455.8
4.0	-0.2313	1198	0.799	51223.7
6.0	-0.2403	967	0.645	49780.5
8.0	-0.2479	837	0.558	49276.4
10.0	-0.2549	759	0.506	49268.5
12.0	-0.2617	717	0.478	49628.4

Table 4: Summary of fitted loading magnitudes.

Factor	Mean $\lvert\lambda\rvert$	Median	$\lvert\lambda\rvert > 0.5$	$\lvert\lambda\rvert > 1.0$
F1	0.665	0.710	349	94
F2	0.084	0.000	30	24
F3	0.130	0.000	34	5

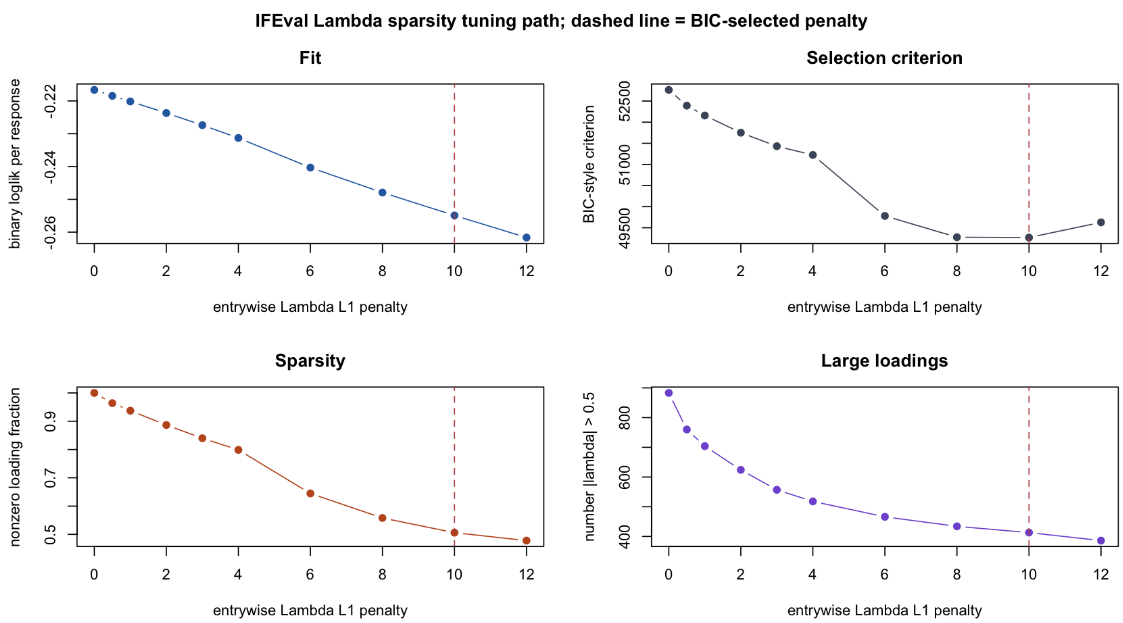

Figure 3: Loading sparsity tuning path. The dashed line marks the BIC-selected penalty.

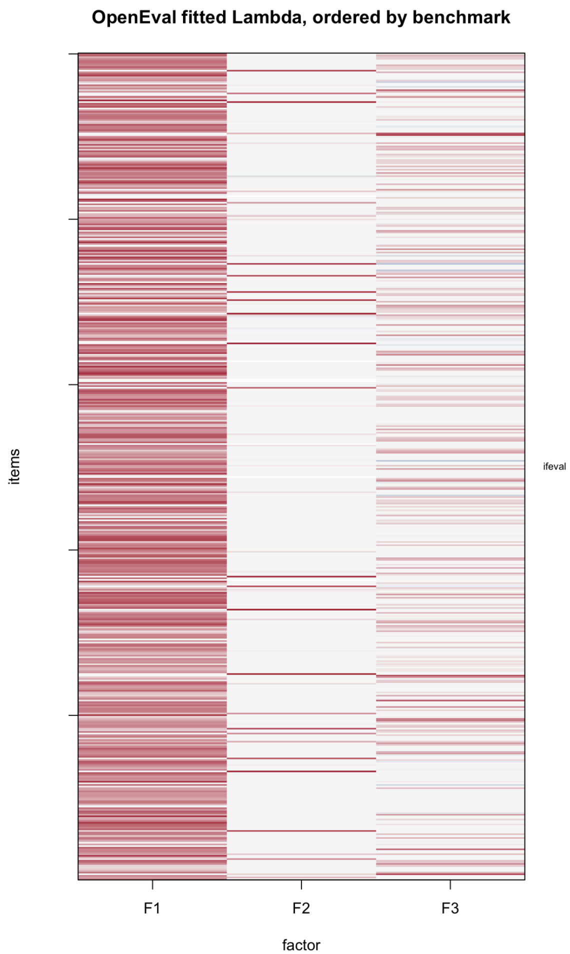

Figure 4: Selected mixture-model loading matrix, ordered by benchmark.

6 Comparison to Ordinary Binary Probit

The ordinary binary probit baseline is fit at the same selected rank, $H = 3$. After signed permutation alignment, its factor space is compared directly with the mixture factor space. This comparison asks whether the predictive and interpretive gains from the mixture model are merely a rotation of a Gaussian factor model, or whether the mixture prior produces a more structured profile geometry.

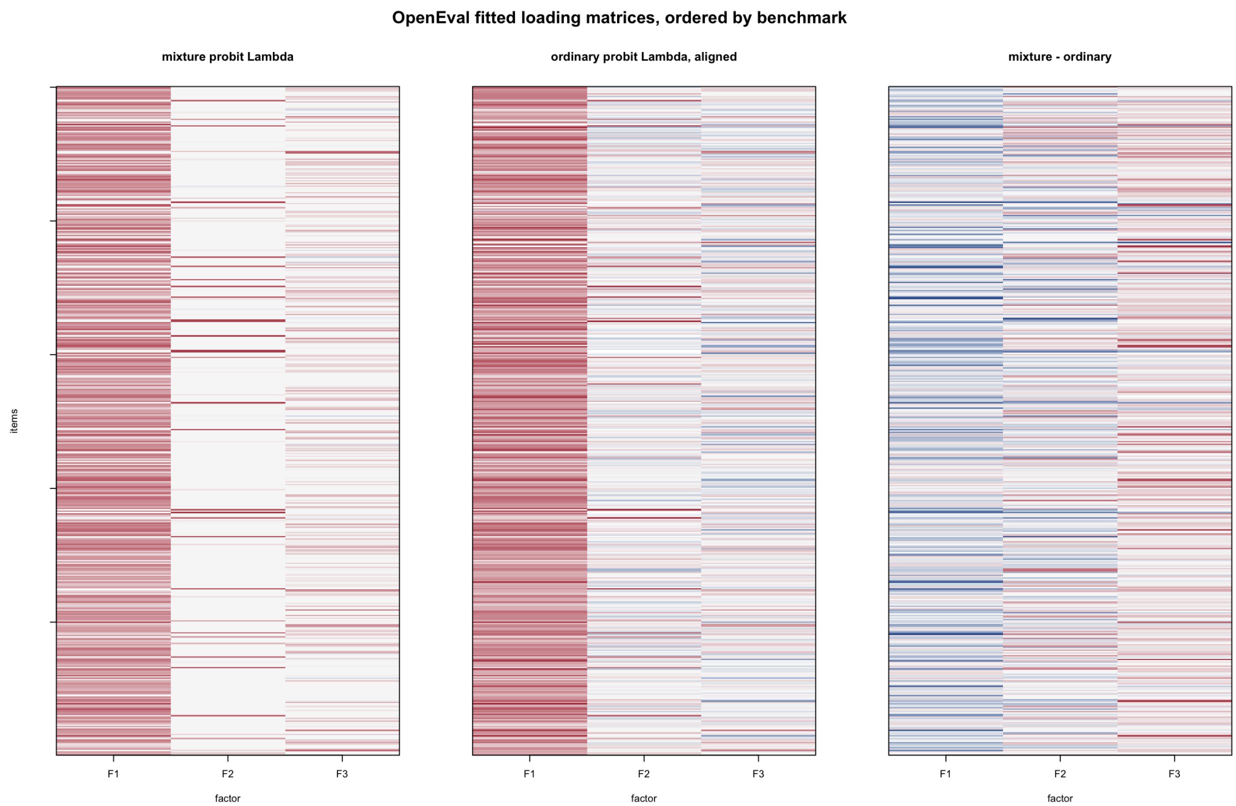

Figure 5: Aligned loading comparison: independent-mixture probit versus ordinary binary probit.

The side-by-side loading plot shows that the broad first-factor structure is shared to some extent, but the mixture model produces a more interpretable partition into marginal mixture profiles. That profile structure is especially useful when reading the factor scores as model types rather than only as continuous latent coordinates.

7 Factor Interpretation

All of the top-loading item sets are IFEval instruction-following prompts, so the factor interpretation is about modes of instruction control rather than benchmark-domain differences. The positive side of factor 1 contains many direct compliance constraints, such as required wrapping, exact ending phrases, paragraph counts, JSON format, capitalization, and no-comma constraints. This factor behaves like a broad general instruction-following axis.

Factors 2 and 3 appear more specialized. Factor 2 has a small number of very large loadings, including prompts requiring strict casing, language restrictions, list or poem formats, and multi-constraint style transformations. Factor 3 loads on a larger but still selective set of items. In prior inspection, these items looked especially connected to creative or generative control: producing songs, poems, essays, stories, or styled rewrites while also respecting formal constraints. The model therefore separates broad compliance from more specific forms of stylistic and creative control.

Table 5: Top-loading item summaries by factor and direction.

Factor	Direction	Mean signed loading	Mean $\|\lambda\|$	Mean item accuracy
F1	Positive	1.196	1.196	0.737
F1	Negative	0.051	0.054	0.600
F2	Positive	0.559	0.559	0.551
F2	Negative	-0.004	0.004	0.773
F3	Positive	0.527	0.527	0.693
F3	Negative	-0.040	0.040	0.777

8 Cross-Loading Structure

The thresholded loading analysis gives 375 factor-only items and 19 cross-loading items at $|\lambda| \geq 0.5$. Among factor-only items, 330 are strongest on F1, 30 on F2, and 15 on F3. All cross-loading items involve F1 and F3, with active pattern 101. This reinforces the view that F1 is broad, while F2 and F3 capture refinements of instruction control that often appear jointly with the broad axis.

9 Mixture Profiles and Factor Visualization

The three marginal mixture coordinates induce a profile label for each model. The highest-accuracy profile is 3-3-3, with 12 models and mean accuracy 0.959. Other high-performing groups include 3-3-2, with 16 models and mean accuracy 0.920, and 3-1-3, with 20 models and mean accuracy 0.898. The profile labels therefore provide a compact summary of model capability across the broad instruction-following axis and the two more specialized axes.

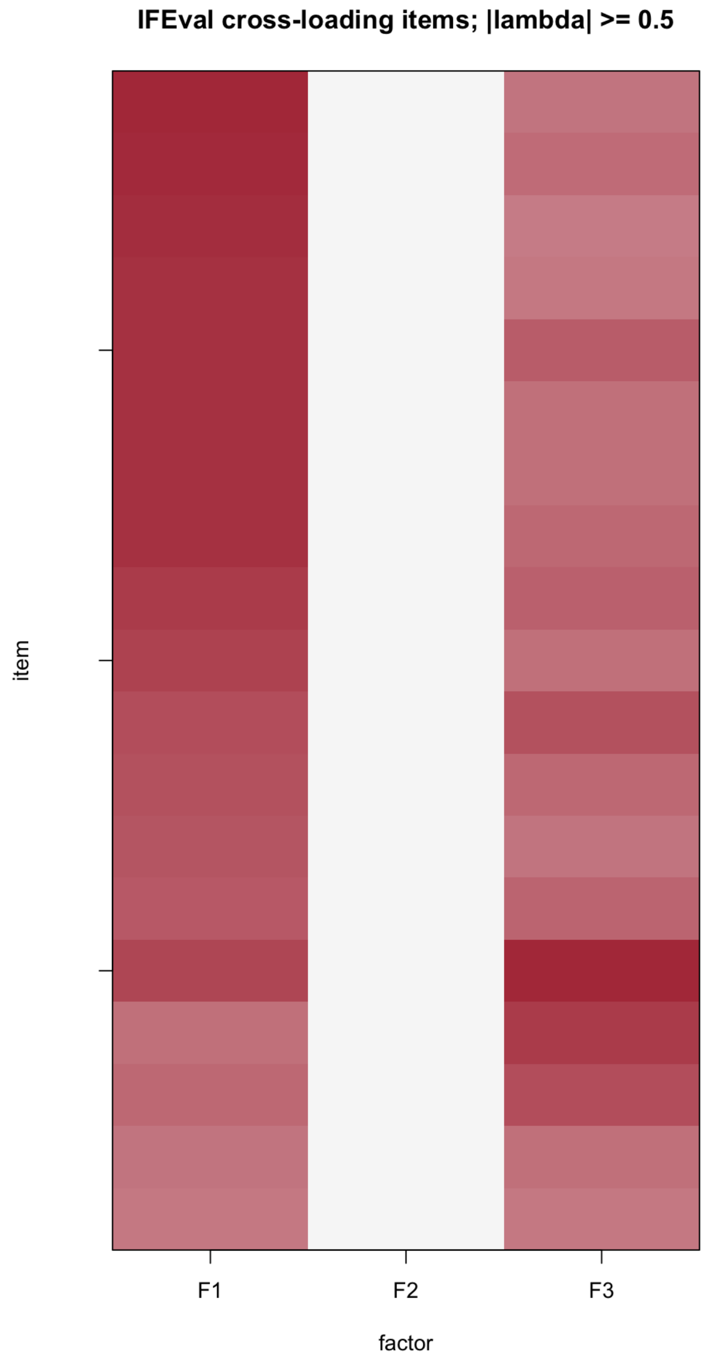

Figure 6: Cross-loading items under the threshold $|\lambda| \geq 0.5$.

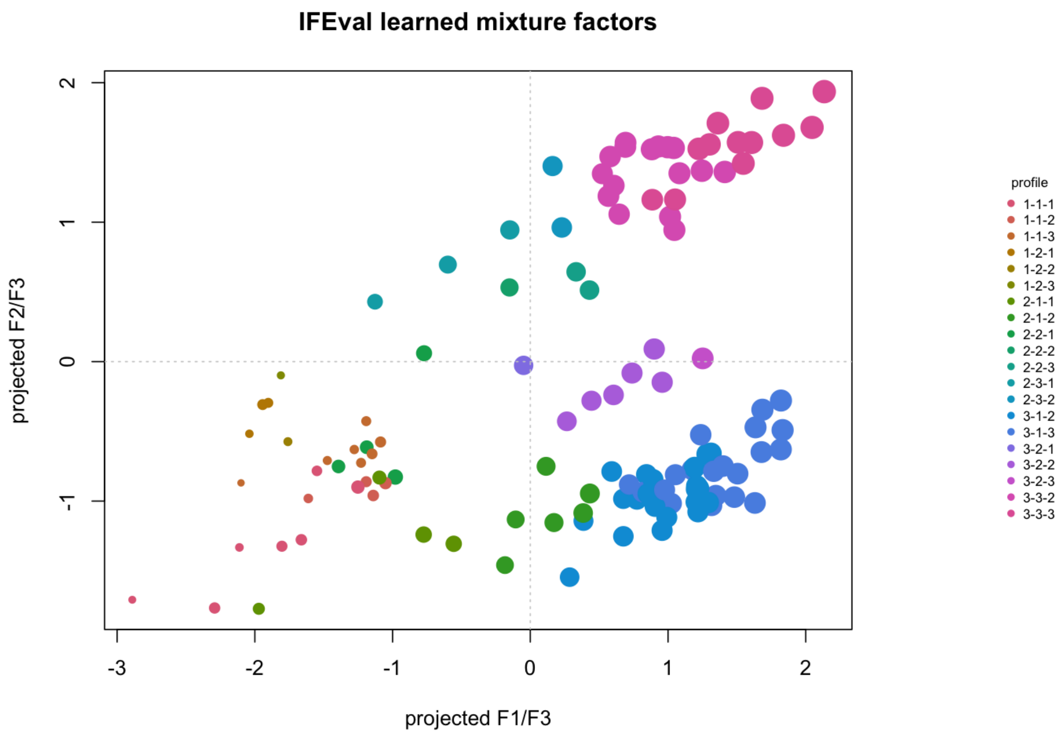

Figure 7: Selected mixture-model factor scores in three dimensions, colored by MAP mixture profile.

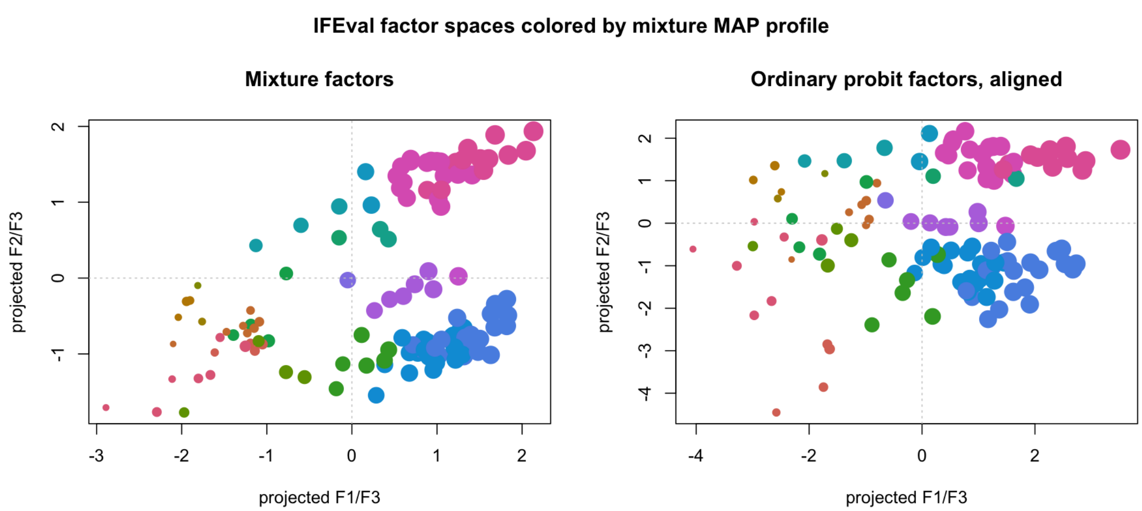

Figure 8: Static three-dimensional projection comparing mixture factors to aligned ordinary probit factors.

10 Summary

The full IFEval pipeline selects a three-factor representation by held-out predictive evaluation, then selects an entrywise loading penalty $\lambda_{\mathrm{L1}} = 10$ by a BIC-style sparsity tuning criterion. At $H = 3$, the $G = 3$ independent-mixture probit fit has the best held-out log score and Brier score among the compared methods, while the ordinary probit factor baseline is close but slightly worse. The selected loading matrix is interpretable: F1 is a broad instruction-following factor, while F2 and F3 capture more specialized modes of constraint satisfaction and creative/stylistic control. The cross-loading analysis shows that many specialized items still involve the broad axis, especially the F1/F3 pattern, which is consistent with a block-like IFEval structure rather than a strict bifactor model. The three-dimensional factor visualization makes this profile structure visible at the model level and gives a useful bridge between predictive evaluation and qualitative interpretation.